

Christopher Barker

Student Developer | Frontend, Applied AI, Data Visualization

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Student developer building frontend, applied AI, data visualization, and interactive web projects with practical implementation evidence. Projects include Python AI pipelines, React/JavaScript interfaces, data analysis workflows, and responsive portfolio systems.

Selected Projects & Evidence

- **Protein AI Pipeline** — Python research pipeline using protein embeddings, image captioning, and similarity evaluation.
- **Pixel Progression** — PHP and JavaScript game system with authentication, save states, leaderboards, and gameplay logic.
- **AI Caption Generator** — Streamlit and OpenAI app for image-aware social captions.
- **HTML Resume Portfolio** — Responsive HTML/CSS portfolio focused on structure, readability, and presentation.
- **Grocery Retail Consumer Analytics** — Retail analytics project using survey data for segmentation and visualization.
- **Opioid Prescribing Risk Analysis** — Healthcare data analysis using public prescribing data and model interpretation.

Education

- **Pacific Lutheran University** — Bachelor of Arts, Individualized Major in Web Development and Interactive Design (2024–2026)
- **Tacoma Community College** — Associate of Arts, Direct Transfer Agreement (2017–2022)

Technical Activities

- **UX/UI Design Intern — TestUnlimited.com**
Converted sketches into UI blueprints and high-fidelity design deliverables. Worked with engineering to implement frontend updates that matched approved designs.
- **Derailed — React, CSS, JavaScript, JSON**
Built a browser game interface with React, JavaScript, CSS, and JSON-based state data. Implemented player movement, collision logic, and gameplay state updates.
- **West Coast Tourism Data Analysis — Python**
Built a Python analysis workflow to compare weekend tourism patterns across retail datasets.
- **AI Protein Sequence Visualization — Python, ESM-3, BLIP-2**
Built a Python AI pipeline that converts protein sequences into visual embeddings and generates image captions for research interpretation.
- **Discord Bot — Python, Git**
Built a Python Discord bot with slash commands, countdown timers, and Git-based version control.

Work & Research Experience

- **Undergraduate Research Assistant — Pacific Lutheran University**

Wrote Python scripts to query biological databases and generate protein structure visuals from sequence data.

- **PC Technician — Pacific Lutheran University**

Troubleshoot hardware and software issues, supported campus users, and mentored new student workers.

Achievements & Awards

- Consortium for Computing Sciences in Colleges (CCSC) — Northwestern Conference (2025)
- Dean's List — University of Washington Tacoma (2024–Current)
- First Generation Fellow — University of Washington Tacoma (2022–2023)
- Outstanding Student Award — Tacoma Community College (2022–2023)

Technical Skills

- **Frontend:** HTML5, CSS3, JavaScript, React, Material UI
- **Applied AI & Data:** Python, Streamlit, OpenAI API, ESM-3, BLIP-2, R, data visualization
- **Programming:** Java, JSON, Node.js, JUnit
- **Tools:** Git, GitHub, Figma, VS Code, Docker, IntelliJ